

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas

Master of Science in Computer Engineering

Aug 2024 – May 2026

Fayetteville, AR

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College

Bachelor of Engineering in Electronics and Communication

Aug 2019 – May 2023

India

TECHNICAL SKILLS

Languages: Python, SQL, C#, C++, Java, C

Data Engineering: Apache Kafka, Apache Flink, Flink SQL, Apache Airflow, ClickHouse, Databricks, Apache Spark, Delta Lake, PostgreSQL, SQL Server, MySQL, Azure SQL, ETL/ELT, data validation

Machine Learning and AI: PyTorch, Scikit-learn, Pandas, NumPy, computer vision, object detection and tracking, LSTM, LangChain, FAISS, RAG, Ollama, Llama 3.2, Streamlit, Flask

Cloud and Tools: AWS Glue, AWS Lambda, Amazon EMR, Amazon S3, Docker, Linux, Git, Jira, Grafana, LabVIEW, NI TestStand

PROFESSIONAL EXPERIENCE

Data Engineering Intern

InsOps Inc.

Jul 2024 – Present

Chicago, Illinois

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant

University of Arkansas, Dept. of Computer Science and Engineering

Sep 2024 – May 2026

Fayetteville, AR

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant

AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

Fayetteville, AR

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer

Soliton Technologies

Jun 2023 – Jun 2024

India

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements

PROJECTS

MiGa - Multi-Chicken Gait Assessment (AICV Lab)

YOLO11n-Pose, OC-SORT, LSTM, PyTorch

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking, and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a detection split of 4,569 frames with 11,422 bounding boxes and a regression split of 55 expert-annotated sequences
- Implemented spatial normalization into local coordinate spaces to eliminate translational variance, achieving keypoint mAP 91.83% at 1,250 FPS, 0.93 MOTA, and 0.91 IDF1

DualFit - Two-Stage Virtual Try-On (AICV Lab)

PyTorch, Res-UNet, flow-based warping, inpainting

- Developed a PyTorch-based two-stage Virtual Try-On model combining flow-based garment warping with Res-UNet image synthesis, with preprocessing and inpainting pipelines to improve garment alignment and visual realism
- Trained and evaluated on the VITON-HD dataset, reaching 24.9 PSNR, 0.913 SSIM, and 5.261 FID